Topic 1 Review · Section 504

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · Tuesday, Nov 17 · Test Wed Nov 18

Name _____ Period _____ Date _____

Directions: This review has the same kinds of questions as the Topic 1 test, in the same order. Show your work. For multiple choice, choose one answer. Finish Part 1 (just like the test) first; Part 2 is extra practice. One question per page. You may use the reference box and a calculator. Ask if you need a question read aloud.

Reference box

Solving: distribute → combine like terms → get the variable on one side → undo + and − → undo × and ÷. Check by substituting.

Same sides (3x + 5 = 3x + 5): infinite solutions · **same x-term, different numbers** (3x + 5 = 3x + 2): no solution.

Inequalities: multiply or divide by a negative → **flip** the sign. < or >: open circle · ≤ or ≥: closed circle.

Slope-intercept form: **y = mx + b** · Standard form: **Ax + By = C** · Triangle area = ½bh · Rectangle area = ℓw

Part 1 · Just like the test (Questions 1–11)

1. Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, y, after x weeks?

A. $y = 15 + 40x$

B. $y = 15x + 40$

C. $y = -15x + 40$

D. $y = -40x + 15$

2. What value of k makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

F. 1

G. 2

H. 4

J. No solution

3. What is the solution set for $3(4 - t) > -2t + 15$?

A. $t > -3$

B. $t < 3$

C. $t \geq 3$

D. $t < -3$

4. Solve the equation shown.

$0.3(4x - 6) = -1.8x + 4.2$

Answer: _____

5. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

Answer: _____

6. Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, y, when Maya buys x pens.

Answer: _____

7. Rewrite the equation in slope-intercept form.

$2x + 5y = 15$

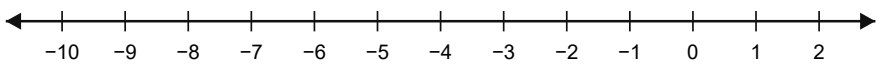
Answer: _____

8. The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for h.

Answer: _____

9. Solve the inequality, and graph the solution on the number line.

$2.5m - 3 < -1.5(m + 18)$



Answer: _____

10. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

Answer: _____

11. A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of n?

Answer: _____

Part 2 · Extra practice (Questions 12–15)

12. Which equation has **no solution**?

F. $3(x - 2) = 3x + 5$

G. $3(x - 2) = 3x - 6$

H. $3(x - 2) = 2x + 5$

J. $3x - 2 = 2x - 3$

13. What is the solution set for $-4x + 7 \geq 19$?

A. $x \geq -3$

B. $x \leq 3$

C. $x \leq -3$

D. $x \geq -6.5$

14. The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for w.

Answer: _____

15. A gym charges a $25 sign-up fee plus $8 per class. Lena can spend at most $90. Write and solve an inequality for the number of classes, c, she can take.

Answer: _____